The space of sociability in the University canteens

Design Thinking study on the innovative commensality

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M.E.N.S.A. GROUP

Mildly Existential Notes on Student Appetites

Overview

What we will talk about in this presentation

1 Introduction



2 Empathise



3 Define



4 Ideate



5 Prototype



6 Test



7 Conclusions





INTRODUCTION

Our *design brief*

AI-Human-Environment collaboration:

opportunities, risks, collaboration models

Our *case study*

The ARDSU canteen system:

From *purely logistical-bureaucratic model* to **emphatic and human-environment-centered** experience

From *neglecting multilingual communication* and *social integration* to a **multicultural and multisensorial organism**

From *spontaneous segregation and alienation among international students* to an **open-minded and pleasant experience**

Starting points

The university canteen is not just a place to eat, but a fundamental **third place** for integration and conviviality among different cultures

The study, focused on the **ARDSU system in Siena**, revealing a significant *communication gap*: despite serving millions of meals, the system lacks multilingual tools to foster genuine interaction between Italian and international students

Our research seeks to transform these spaces from mere logistical hubs into **environments of active cohabitation**, balancing local identity with the diverse needs of a global student population



EMPATHISE

THIRD PLACE

- Canteen as a *third place* where social classes are overthrown (Oldenburg, 1989)
- *Café of connection*: a catalyst against isolation (Makhathini & Kanosvamhira, 2025)
- *Food choice mimicry* (Gligorić, 2024)
- The importance of *commensality* and the risk of loneliness (Dunbar, 2017; Mensah & Tuomainen, 2024)
- Digital intrusion: *fragmented togetherness* (Zhang, 2025) and the problem on the representation of ethnic meals (Dong, 2025)

MULTISENSORIALITY

- *Gastrophisics* (Langella et al., 2024): the perception of taste is influenced by a multitude of environmental cues
- *Embodied cognition* (Pan et al., 2025): the environment plays a crucial role in the experience

AI IN THE CANTEEN

- Efficiency on the service (Wang & Wang, 2026)
- LLMs: personalization with attention on food allergies and cultures (Gavai, 2025; Wang et al. 2025)
- Ethical questions: directness and explainability (Yang et al., 2024)

Desk Research

This phase has helped us to reimagine the canteen as a holistic system centered on:

- **Social connection** to discourage isolation
- **Multisensory design** to enhance meal quality and well-being
- **Ethical AI integration**

User Research

This phase focused on understanding how students experience the canteen in real-life situations

We applied several *empathic methods*: these tools allowed us to observe behaviors, emotions and difficulties directly in context

SHADOWING

We followed international students throughout their entire canteen journey

Menu navigation and staff interactions, combined with time pressure, trigger **confusion** and **stress** for users

SERVICE SAFARI

Crowding and *speed* negatively impact choices, whereas *convivial moments* elevate the overall user experience

By **experiencing the canteen as users**, we documented each step of the service process

WHAT/WHY/HOW

This method helped us analyze *critical incidents* such as wrong orders, misunderstandings, and embarrassment.

These situations revealed **systemic problems** related to communication, service design and staff workload

CULTURAL PROBES

Participants documented *smells*, *sounds*, *comfort*, *lighting*, and *social atmosphere*

While **Wi-Fi** and n**atural light** are strengths,
noise, **overcrowding** and **missing multicultural elements** remain critical gaps

Beyond inconvenience: language barriers are functional obstacles affecting student confidence, participation, and well-being

Speed vs. Interaction: The chaotic queue generates social pressure, discouraging students from asking questions and inhibiting inclusion

The cost of exclusion:

- *Imitation*: Choosing what others eat to avoid questions
- *Isolation*: Eating alone to escape social stress
- *Avoidance*: Abandoning the canteen altogether

Seed insights



*The canteen serves its purpose but **fails to feel welcoming**. We must address the **social** and **emotional barriers**, not just the logistical ones*

*Beyond language, **emotional** and **social barriers** limit student participation.*
*These invisible factors **are critical** to student confidence and well-being*



DEFINE

The stakeholder map

Stakeholders:

-Erasmus students

-Italian students

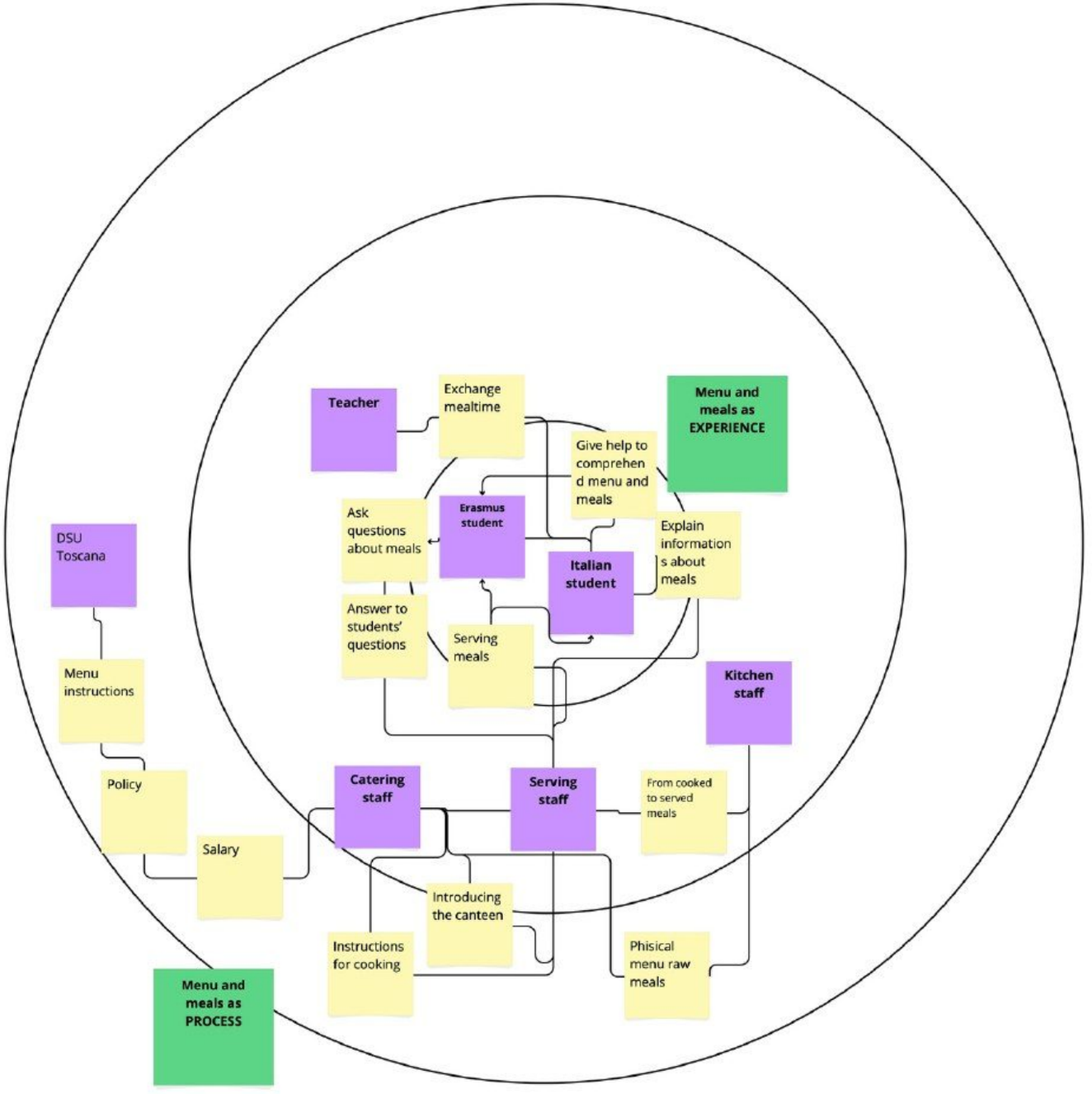
-Kitchen staff

-Serving staff

-Catering staff

-Teachers

-DSU Toscana



Point Of View

The POV analysis synthesizes stakeholder goals and needs, establishing a solid foundation for idea generation



Students' POVs

divided among different student-stakeholder

- **On-site student**: Healthy meals to power through full days of classes
- **Out-of-town student**: A comfortable environment and eco-friendly food options
- **Commuter student**: Quick, convenient snacks, prioritizing speed over quality
- **International student**: Familiar flavors and a service free of language barriers
- **Special-dietary-need student**: Clear, instant information to ensure dietary autonomy



Staff's POVs

divided among two different type of staff

- **Senior staff**: Challenged by technological adoption and communication barriers
- **General personnel**: Driven by service speed and workplace social connections



Faculty members' POV

divided among two different type of lecturers

- **Regular faculty**: Needs a balance of social engagement and time-efficient dining between commitments
- **Occasional faculty**: Seeks a quiet, solitary environment for a fast and functional meal

Emphatic personas



Sandra
out-oftown student



Nora
Erasmus student



Joanna
Celiac student



Alex
Vegan student



Giacinto
Commuter student



Giovanni
Food serving assistant



Luca
Student living in town



Leonardo
Daily canteen user teacher



IDEATE

How Might We...?

HMW questions were formulated to **transform the critical issues** that emerged from the research into **concrete design opportunities**, structured around four fundamental pillars

Environmental quality

How might we transform the cafeteria into a social hub by improving sensory elements (like acoustics and color) and using feedback systems to monitor satisfaction?

Student–staff communication

How might we improve dialogue between staff and students using simple tools that reduce workload and foster inclusion, while ensuring mutual understanding without replacing human interaction?

Operational efficiency

How might we balance speed and service quality by streamlining orders, offering fast-track options, and using digital tools to reduce hesitation and allow customization?

Multiculturalism and health

How might we develop inclusive menus that respect cultural and dietary needs, while simplifying communication and encouraging the exploration of international cuisines?

Design challenge

A set of design challenges is emerged, structured around three areas

AI for inclusion

- Smart tables with a real-time translation assistants for international students.
- Generative AI: for personalize dining and cultural adaptation

Digital flow management

- *Smart turnstiles*: Real-time attendance and queue reduction.
- *Schedule sync*: Linking meal reservations to class timetables.
- *Feedback loop*: Post-meal evaluation via digital tickets

Environmental redesign

A multicultural lounge featuring odor-resistant walls and adaptive audio-visual systems to resolve structural noise and odor issues



PROTOTYPE

Meet Jens
the solution

A multilingual and cultural assistant for university canteens

Supporting international students understanding canteen behavior and interactions.



Hello, my name is Jens. I'm you local language buddy.

I'm here to help you understand the university canteen. Translate ingredients into your language, explore the flavours and culture of our country and university.

Feel free to ask me questions about words you don't understand. I'll make sure to give you friendly, useful and personalized information

Personality

- Helper
- Friendly
- Empathetic
- Approchable
- Culturally sensitive

Tone

Informal	Formal
Humorous	Serious
Adaptive	Technical
Personal	Impersonal

Main Functions

Jens acts as a linguistic and cultural assistant in the university canteen.
His main functions are:

- Real-time translations
- Explanation of menus, ingredients, and allergens
- Support in situations of misunderstanding or uncertainty
- Cultural mediation and contextualization of dishes and habits
- Promotion of daily language learning

- Reduction of anxiety during queues and interactions
- Encouragement of user autonomy
- Stimulation of social and multicultural interaction

Goal

It's goal is not to replace human staff, but to support communication, improve user experience, and strengthen cultural inclusion in the university canteen.

TODAY

Hello and welcome. Ask me specific questions, I will help you find the information you are looking for. 14:04

who are you? 14:04

Sono Jens, un assistente virtuale della mensa universitaria. Aiuto gli studenti a comunicare e capire meglio l'italiano. 14:04

what can you do? 14:04

Posso aiutarti a capire il menu, tradurre parole, spiegare piatti e rispondere a domande culturali. 14:04

What is today's menu (mondaly) 14:04

Lunedì, i primi piatti sono: pasta al pomodoro e risotto alle verdure. 14:04

how can I ask for desert? 14:05

AI You can say: "Posso avere un dolce, per favore?" 14:05

And if I want to say "bread"? 14:05

In italiano "bread" si dice pane. 14:05

Write here

Chat & Vocal assistant



Write here



TEST

The evaluators

eight in total, three international and five Italian

I like I wish What if (I session)

I like

"The aesthetics. Looks real. Is fast and creates engaging interactions."

I wish

"A more long-lasting and practical function."

What if

"We included a weekly menu translation or a link to the ARDSU website?"

I like I wish What if (II session)

I like

"The breadth knowledge and empathy."

I wish

"For fully integrated system using QR codes in the canteen."

What if

"We shouldn't have to explain habits that should largely be taken for granted?"







CONCLUSIONS

Main goals of the research



The **literature analysis** provides an international perspective open to innovation in dining spaces

Empathy-driven methods highlights the harsh realities and significant growth opportunities for international students in Italy

Jens serves as an effective tool to rethink social dynamics and re-evaluate the needs of canteen users in a pragmatic and engaging way

Main limitations of the research



The **limited number** of students involved in the evaluation

The i**ncomplete development** of the digital human's knowledge base

The **lack of involvement** of other stakeholders (staff and faculty members) and extreme users

The absence of prototype integration into a **digital system**

Final insights

- ***Goal***: Transforming ARDSU canteen into empathetic and intercultural hub
- ***Approach***: Focused on ethical and *human-environment-centered* innovation
- ***The solution***: *Jens* reduces communication barriers and social isolation
- ***Results***: User evaluations show highly promising feedback
- ***Impact***: Generated creative and engaging food-centered learning ideas
- ***Next step***: Integrate the digital human into real-world systems
- ***Collaboration***: Involve all stakeholders in further development
- ***Vision***: Expand multisensory experiences to promote inclusive student life

Thank You

Grazie *Gracias*

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